

UNIVERSITY FINAL PROJECT DEFENSE

AI-Powered CV Pre-Screening Assistant Using Semantic Search & LLMs

Accelerating Recruitment Through Intelligent Candidate Matching

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Role: Senior Solution Architect & AI Lead | Department of Computer Science

SPEAKER NOTE:

Welcome honorable examiners and lecturers. Today I present my final project on an AI-powered CV Pre-Screening Assistant, leveraging semantic search and LLMs to solve critical talent acquisition bottlenecks in IT consulting.

PART 01

Context & Problem Statement

Examining the core recruitment bottlenecks in modern IT consulting

SPEAKER NOTE: In this first section, we examine the current operational environment of our IT consulting company and the severe friction caused by high-volume manual candidate screening.

| Company Context & Application Volume

IT Consulting Dynamics

The company specializes in providing top-tier software engineers to enterprise clients. Business success relies directly on speed-to-market and high candidate matching quality.

Delayed staffing directly impacts client project timelines and corporate revenue.

The High-Volume Challenge

Every single job vacancy receives between 100 to 500+ CV submissions across multiple platforms. Recruiters struggle under immense manual processing loads.

Navigating hundreds of unstructured PDF/Word resumes creates a critical operational bottleneck.

SPEAKER NOTE:

As an IT consulting firm, our core asset is engineering talent. However, receiving hundreds of applications per role slows down recruitment, directly affecting client satisfaction when roles stay open too long.

| The CV Screening Bottleneck

Operational Friction Points

- **Repetitive Manual Tasks:** Reading hundreds of formatted CVs line-by-line.
- **Fatigue & Inconsistency:** Evaluation criteria shift as fatigue sets in across long screening hours.
- **Overlooked Talent:** Highly qualified candidates are missed due to keyword mismatch or human error.

SPEAKER NOTE:

Manual review of 100+ CVs leads to recruiter fatigue, subjective evaluation, and lost talent. Our goal is to streamline this pipeline through intelligent vector indexing.



I Quantifying Recruiter Workload

85%

Screening Time Spent Reading Resumes

The Cost of Manual Processing

- Recruiters spend an average of **40 to 60 hours per vacancy** just reviewing initial applications.
- Less than **15% of total time** is available for high-value interview and candidate engagement tasks.
- Sluggish screening creates candidate drop-offs and delays client delivery by up to 3 weeks.

SPEAKER NOTE: Up to 85% of recruitment time is spent on initial filtering. By reducing this manual overhead, recruiters can focus on high-touch interviews and strategic talent engagement.

PART 02

Traditional vs. Proposed System

Evolving from syntactic keyword search to semantic intelligence

SPEAKER NOTE: Next, we compare legacy Applicant Tracking Systems (ATS) and manual methods against our modern vector-based LLM solution.

| Existing System & Limitations

✖ Traditional ATS Approach

- **Exact Keyword Match:** Searches for exact string literals (e.g. "React Native").
- **Manual Excel Comparisons:** Copying candidate skill tags into spreadsheets.
- **Zero Semantic Context:** Misses related expertise if terminology differs.

⚠ Core Limitations

- Fails when candidates express experience using contextually equivalent terms (e.g. "Mobile Dev" vs "Flutter Engineer").
- Rigid rule filters reject strong candidates due to minor formatting discrepancies.
- Completely unscalable for high-growth enterprise hiring needs.

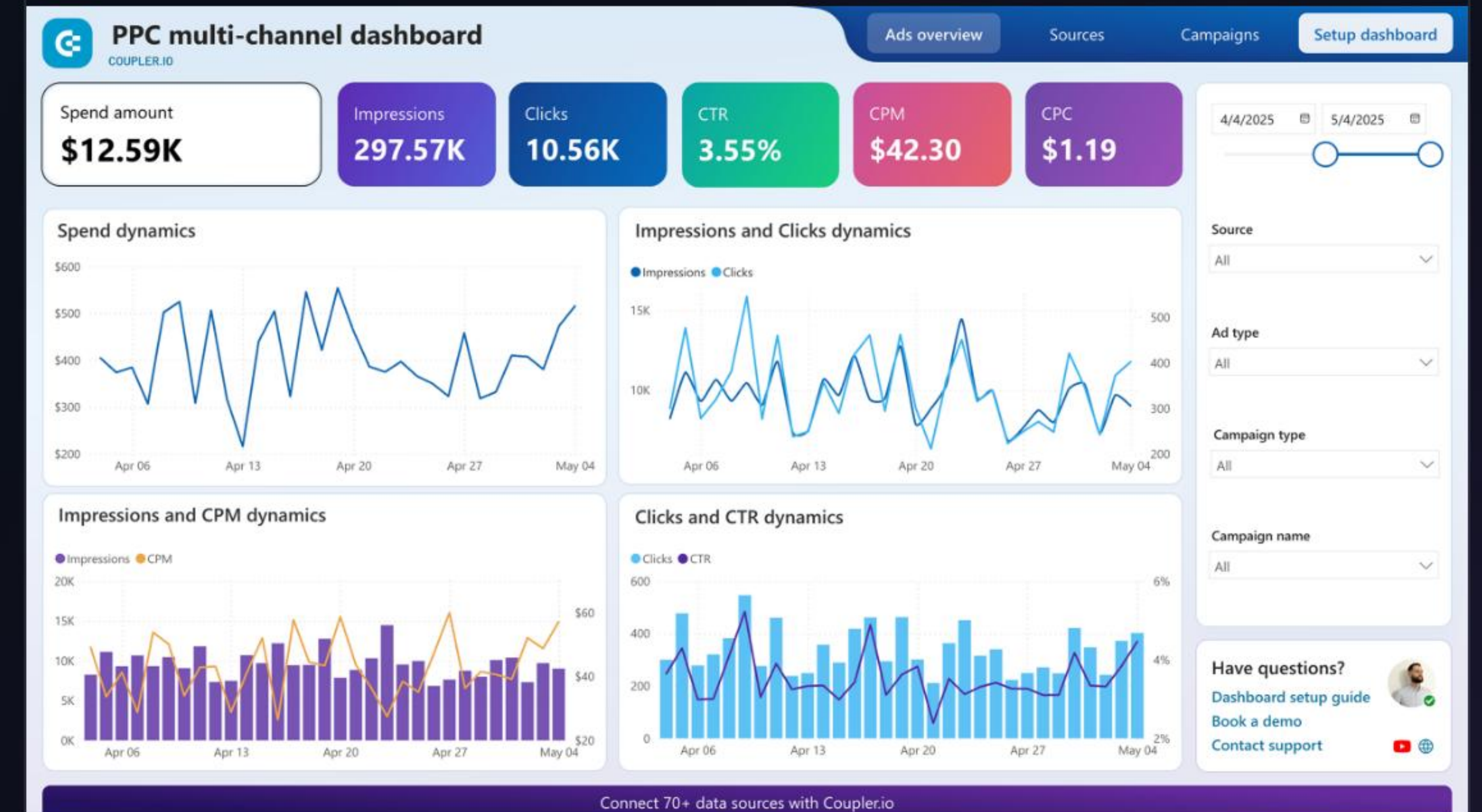
SPEAKER NOTE: Traditional ATS relies on exact string matches. If a job calls for "Mobile Developer" and a resume says "iOS & Android Engineer", legacy keyword filters often fail completely.

Proposed Solution: AI Assistant

Semantic Intelligence & LLM Synthesis

- **Automated Structuring:** Parses raw unstructured PDFs into JSON objects using Pydantic.
- **Vector Search:** Computes high-dimensional dense embeddings to perform similarity scoring.
- **LLM Reasoning:** Generates transparent, natural language explanations for rankings.

👤 **Human-in-the-Loop:** Recommends candidates; hiring decision stays with recruiters.



SPEAKER NOTE:

Our proposed solution automatically parses, embeds, and ranks resumes based on semantic intent, using LLMs to provide concise reasoning while keeping human recruiters in full control.

| Core Objectives & Metrics



Reduce Workload

Cut initial screening time by 75%, allowing recruiters to process hundreds of candidates in minutes.



Standardize Reviews

Eliminate subjective bias and fatigue through objective high-dimensional vector similarity scoring.



Improve Matching

Capture latent skills, related frameworks, and experience context beyond surface-level keywords.

SPEAKER NOTE: The core objectives center on three pillars: drastically reducing screening duration, establishing standardized evaluation metrics, and elevating overall candidate match quality.

PART 03

Technical System Architecture

Deep dive into the microservices, embeddings, and vector database engine

SPEAKER NOTE: Moving to the system architecture section, I will explain the data flow, software stack, and mathematical concept behind high-dimensional vector embeddings.

| End-to-End System Components



Frontend Application

Flutter Web/Desktop UI

Recruiter interface for uploading CVs, conducting natural language searches, and visualizing rank scores.



API Backend Engine

FastAPI & Python 3.11

Asynchronous REST API orchestrating document parsing, schema validation, and pipeline processing.



Vector Store & LLM

ChromaDB + Ollama/OpenAI

Stores candidate embeddings and runs local/cloud LLMs for ranking justification.

SPEAKER NOTE:

The system architecture cleanly decouples the Flutter recruiter interface from the FastAPI backend microservices, connected to ChromaDB for vector retrieval and LLMs for contextual explanations.

| End-to-End Data Pipeline



SPEAKER NOTE: This pipeline shows how raw candidate CVs travel from initial PDF parsing to structured JSON, dense embedding calculation, ChromaDB indexation, and fast cosine similarity lookup.

| Vector Search vs Keyword Search

Keyword Match Failure

Query: *"Mobile developer experienced with Dart and Riverpod"*

- ✗ Misses: "Flutter Engineer"
- ✗ Misses: "Cross-Platform Mobile Developer"

Semantic Vector Match

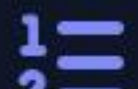
Embeddings project concepts into high-dimensional vector space.

- ✓ Matches: "FlutterLead" (0.91 similarity)
- ✓ Matches: "Mobile Software Engineer" (0.87 similarity)

SPEAKER NOTE:

Dense vector embeddings map words into semantic vector space. Even if a resume omits the exact query words, closeness in vector space correctly identifies related engineering skills.

| Recruiter Functional Features

-  **Multi-Format Bulk Upload:** Batch processing of unstructured PDF and DOCX resume files.
-  **Natural Language Queries:** Search candidates using plain English job requirement descriptions.
-  **Automated Rank Indexing:** Real-time distance scoring sorted by highest semantic alignment.
-  **Transparent AI Explanations:** Concise LLM-generated summaries highlighting specific candidate skill alignment.

SPEAKER NOTE: Key features designed specifically for recruiters include bulk resume uploads, natural language query search, dynamic candidate ranking, and transparent AI reasoning cards.

| Key Business & User Benefits



For Recruiters

- 75% faster pre-screening.
- Reduced manual data entry fatigue.
- Consistent objective evaluations across candidates.



For Consulting Firm

- Rapid placement of qualified talent.
- Higher client satisfaction & trust.
- Scalable recruitment operations without head-count growth.



Cost Efficiency

- Low operational footprint using open-source local LLMs & ChromaDB.
- Reduced recruitment cost per hire.

SPEAKER NOTE: The business impact directly targets core KPIs: faster candidate turnaround for clients, enhanced recruiter productivity, and reduced cost per hire.

| Limitations & Future Roadmap

Current Limitations

- Dependent on candidate CV completeness and document text formatting.
- Cannot evaluate soft skills or culture fit without live interviews.
- Requires human recruiter validation for final hiring compliance.

Future Enhancements

- Direct ATS API integrations (Greenhouse, Workday).
- Automated interview question generation based on CV skill gaps.
- Feedback learning loops to refine similarity weighting over time.

SPEAKER NOTE: We maintain a realistic evaluation of current boundaries while outlining an exciting future roadmap including direct ATS integration and automated interview preparation.

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Thank You!

Questions & Defense Discussion

Candidate: Dika Wardani | AI-Powered CV Pre-Screening Assistant

SPEAKER NOTE: Thank you for your time and consideration. I am now open to questions from the board of examiners and faculty members.

Image Sources



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